

Accessible PDFs with L^AT_EX

for ~~DUMMIES~~ Teachers

Bryan Lewis and Carla Zeigler, Garrett College, January 2026

Why L^AT_EX?

- L^AT_EX is the de facto standard for the communication and publication of math and science. Chances are you *will* encounter L^AT_EX in STEM!
- L^AT_EX is widely used in publishing because it clearly separates presentation from content.
- L^AT_EX typesets technical symbols and mathematics beautifully.

Why Accessibility?

The Department of Justice established **April 24th, 2026** as the compliance deadline for the updated Title II regulations of the Americans with Disability Act.

All course materials in Canvas, Blackboard, and other LMS platforms must be accessible to students with disabilities.

Regulations require compliance with WCAG 2.1 AA Accessibility Standards.

Why Accessibility?

The real reason: **It's the right thing to do!**

Lots of us have difficulty reading/understanding/processing digital media. We must strive to make concepts presented digitally accessible to everyone. Difficulty reading or understanding digital content is an unnecessary barrier to learning.

Accessible PDF Output is (Recently) Possible!

- ...Thanks to *years* of diligent work by the L^AT_EX project developers. See this presentation at PDF Days 2025 Berlin:

<https://www.latex-project.org/news/2025/10/30/pdfadays>

- Also check out the many solution posters from the conference:

<https://pdfa.org/the-winning-technical-poster-at-pdf-days-europe-2025>

- And this excellent reference from Richard Wong:

<https://richardwong.rice.edu/LaTeX-Accessibility-Primer/>

- Our slides walk you through a few basic steps to get things working.

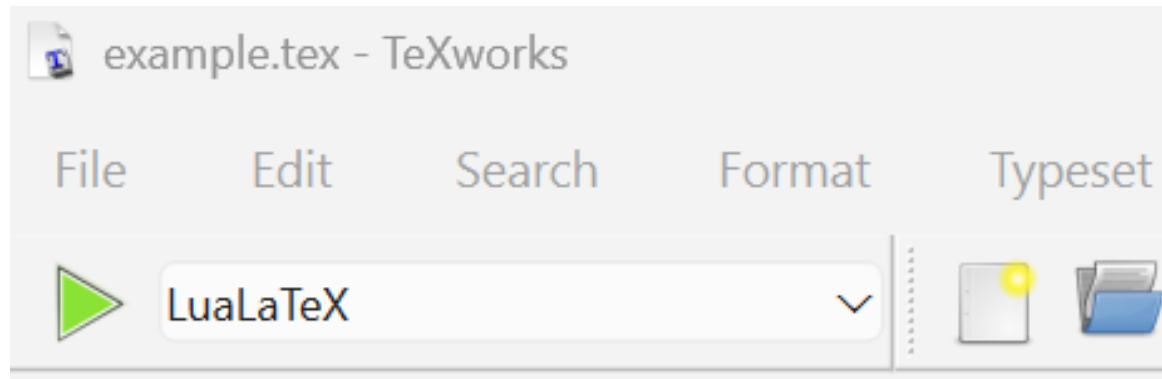
Step 1: Use `lualatex`

- $\text{\LaTeX}$ can be processed by many different rendering engines (`pdflatex`, `lualatex`, ...)
- `lualatex` is currently the best engine to use for accessibility.
- On many GNU/Linux systems simply install with:

```
sudo apt install texlive-luatex
```

Using lualatex with MiKTeX on Mac and Windows Systems

1. Install MiKTeX (<https://miktex.org/download>)
2. Open the installed TeXworks editor and select the LuaLaTeX engine from the drop-down menu (see figure below).

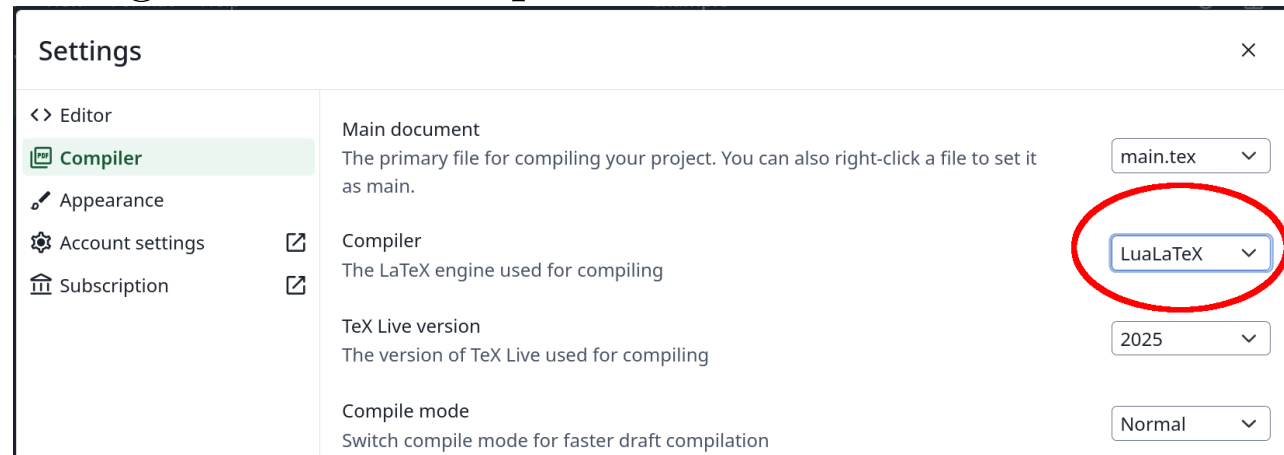


That's it! Note that MiKTeX may need to download and install a few packages the first time you select the LuaLaTeX option.

Using lualatex with Overleaf

Overleaf is a popular online $\text{\LaTeX}$ editor with over 20 million users. You can set it up to use the `lualatex` engine. Alas, this might only work with the paid version ☹.

- From the File menu, select the “Settings” option.
- Navigate to “Compiler” and select “LuaLaTeX” as shown in the figure below.



- See <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>.

Step 2: Metadata in your documents

Add the following to the very top of your document (adjusting `lang` as needed):

```
\DocumentMetadata{
  tagging=on,
  tagging-setup={math/setup={mathml-SE,mathml-AF},math/alt/use},
  pdfstandard=ua-2,
  pdfstandard=a-4f,
  lang=en-US
}
```

This turns on tagging and embeds two kinds of mathematical notation markup for widest compatibility with reader software *and* adds LaTeX-based alt-text tags to images for older readers that can't understand MathML.

PDF document metadata

Add the following to your preamble above `\begin{document}` (adjusting entries as needed):

```
\usepackage{hyperref}
\hypersetup{
  pdftitle={Accessible PDF with LaTeX},
  pdfauthor={Bryan W. Lewis},
  pdfsubject={Accessibility},
  pdfkeywords={Accessibility, LaTeX, PDF},
}
```

Use the `unicode-math` math symbol package

Include the following package in your document preamble:

```
\usepackage{unicode-math}
```

Avoid use of the following packages if possible (`unicode-math` may work in their place):

~~\usepackage{amsmath}~~

~~\usepackage{amssymb}~~

This may not be strictly necessary but tends to produce good embedded MathML output.
BREAKING NEWS, as of Spring, 2026 amsmath should work OK!

Be sure to add text descriptions to every included image

Whenever you include an image in your document, be sure to add a text description with `alt`.
For example:

```
\includegraphics[alt={Image description}, width=\textwidth]{filename}
```

Congratulations! Your L^AT_EX-generated PDF documents are now accessible!

Documents prepared in this way should score 100% in Blackboard Ally's checker.

See

<https://github.com/bwlewis/latex-pdf-accessibility/tree/main/example>

for simple example documents that you can use as starting points.

Setting up NVDA reader software with MathCAT

- The reader software with the most complete support is NVDA (NV Access) equipped with the MathCAT add-on. It's free, supports audible and braille output in many languages, and works very well!

<https://www.nvaccess.org/download/>

- Install NVDA from the above link...

- Bring up the NVDA menu and select “Add-on store...” from the “Tools” menu.
- Right-click on the “MathCAT: speech and braille from MathML” add-on and select “Install” You will see “Downloaded, pending install” in the Status column. (This can take a while.)
- You’ll be asked to re-start NVDA for the add-on to take effect.

You’re all set to read beautiful PDF documents with $\text{\LaTeX}$ math aloud or in braille!

We recommend using Microsoft Edge to view PDF documents to be read by NVDA—it worked fine for us! Firefox also worked. Weirdly, Google Chrome’s PDF viewer didn’t work well with NVDA on our Windows 11 computer.

As an alternative, Adobe Acrobat Reader should also work.

Issues

- These techniques are all *very* new! Most (all?) accessibility checkers don't completely work yet. That includes, at least to some extent, Blackboard's Ally built-in checker as of January 2026 (which only understands alt-text equation image tags and does not understand embedded MathML yet).
- *Most* L^AT_EX packages “just work.” Your mileage may vary however with less mainstream packages. If you run into issues you might need to consider package changes.
- As mentioned in the last slide, choice of PDF viewer and reader compatibility can be finicky. We expect fewer glitches as the software matures.

Old PDFs

Many of us are stuck with old PDFs without source code. Carla suggests using AI to transcribe them if all else fails. This actually can work pretty well!

We suspect it's only a matter of time before screen readers start to incorporate that capability.

About us



Carla Zeigler is the Coordinator of Distance Learning and Instructional Design at Garrett College.



Bryan Lewis teaches math at Garrett College and has been using $\text{\LaTeX}$ since the 1990s. He's written dozens of journal papers, a book or two, many talks, and lots of classroom notes in $\text{\LaTeX}$.

Slide source code, examples, and more

These slides, examples, and example videos are available from:

<https://github.com/bwlewis/latex-pdf-accessibility>

and can be freely used and redistributed without restriction.